Chapter 4
Updates And Stock Count Procedures

Regular Updates And Procedures

/STINT

In order to improve the speed and quality of information displayed within the Inventory Management application a series of weekly and periodic balances are maintained. These balances need to be updated at week, month and year end.

At month end, both week end and month end should be run consecutively.

At year end, week end, month end and year end should be run consecutively.

The month end run normally includes a valuation of stock movements and this, therefore, is prompted.

These batch functions are accessed by selecting the option Updates and Stock Taking Procedures. They are described in detail as follows:

Week End Update

1/STINT

This batch function performs the following tasks:

- If the usage cycle set at company level is W, for weekly, then average weekly usage will be recalculated for each product/stockroom.
- The physical stock at the end of the week is stored in a separate field 'Stock at start of week' for each product/stockroom.
- Usage this week is set to zero for each product/stockroom.
- The current week number on the calendar file is incremented by 1. If the end of year is reached then week number is set to 1.

Note: If processing the last week of the year, this update should immediately be followed by month and year end processing since the year number on the Calendar file is only increment by the year-end update.

To process the week end update, simply take the option to submit the job. No selection parameters are required.

Month End Update

2/STINT

This batch function performs the following tasks for each product/stockroom.

- If the usage cycle set at company level is M, for monthly, then the average monthly usage will be recalculated.
- Usage this period is set to zero.
- Number of sales orders this period is set to zero.
- Number of substitutions this period is set to zero.
- Current physical stock is stored in a separate field 'Stock at start of period.'
- The stock value is calculated using the appropriate costing method and stored in a field called 'Stock value at start of period.'

The function will also:

- Increment the current period number by 1. If the end of the year is reached the period number is set to 1.
- Produce a stock movements valuation report if this has been requested as part of the submission process. This report is produced based on parameters entered at that time.

On taking the option for month end update, a screen is displayed requiring the parameters for the stock movements valuation report. Enter the parameters and press **F8** to accept, or press **F3** if you have already produced this report for this period. (See the section on Reports for a full explanation of this report).

Year End Update

3/STINT

This batch function performs the following tasks:

- Allows the number of weeks this year and last year specified on the calendar to be updated.
- Allows the number of periods specified on the calendar to be changed.
- Sets any yearly accumulators to zero.
- Increments year number by 1.

The calendar file maintenance screen will be displayed. Review and update the number of periods and the number of weeks this year and last year so that they reflect the new year.

When you press **Enter** a batch job is then submitted to perform the year end update.

Reorder Recommendations

The investment in Inventory typically represents one of the largest single areas of investment in a business, often as high as twenty-five percent of total assets. It is, therefore, vital that planning and control of its acquisition and disposition are exercised.

A number of techniques have been developed to assist in this control. In the sphere of a manufacturing operation where demand is considered to be 'dependent' the most common forms of control are: Material Requirements Planning or Just In Time. These techniques work in the relatively predictable environment where demand is dependent on an agreed production schedule. Such control techniques are provided in the Manufacturing Applications of Style.

However, the demand for end product parts is primarily influenced by factors that are independent of company decisions. These external factors include a certain amount of random variation in the demand for such parts. As a result, forecasts of demand for these parts are typically projections of historical demand patterns to determine average usage. The concept of independent and dependent demand are important in selecting appropriate inventory management techniques.

There are generally considered to be two types of Inventory: Movement and Organisational.

Inventory Holding

Movement Inventories

These exist purely because time is required in which to make and transport products. The most obvious of these is work in progress for a manufacturing environment and stock in transit for a distribution environment.

Organisation

Inventories exist to decouple successive stages of production and distribution. Finished goods inventory makes it unnecessary to produce directly to meet the demands from distribution centres. This permits flexibility in scheduling manufacturing and assembly operations. Organisational inventory can be held to meet different requirements; these should also be understood and are: cycle stock, safety stock and anticipation stock.

Cycle Stock

Is a result of buying or making more than is needed to fulfil the immediate requirement. The one time costs of raising an order and perhaps setting machines or transport costs must be set against the costs of holding stock, to determine the 'Economic Order Quantity.'

Safety Stock

Is to protect against uncertainty. That is not only the uncertainty of demand, but also the likelihood that planned lead times will be achieved.

Anticipation Stock

Is that held when demand exceeds the rate at which the product may economically be supplied. This particularly relates to Seasonal products.

Inventory Decisions

The two decisions made in inventory are how much to order and when to place the order. Four main factors influence the decisions, these are:

1. Forecast demand.
2. Replenish lead time.
3. Inventory related cost.
4. Management policies.

Two important parameters which assist you in making these decisions are:

- Economic Order Quantities (EOQ).
- Economic Time between Orders (ETBO).

The standard calculation of EOQ's:

$$EOQ = \sqrt{\frac{2CPxAU}{CH}}$$

where CP=Cost of order placement

AU=Average usage per annum

CH=Inventory holding cost per year

$$= \frac{\text{Item Cost} \times \text{Carrying Rate}}{100}$$

Once EOQ has been determined then:

$$ETBO = \frac{EOQ}{WU}$$

where WU=Average usage per week

The resulting ETBO is in weeks.

Style Inventory Management provides a range of facilities which contribute to the management of independent demand products which are traditionally controlled by reorder point techniques.

- A value/usage matrix enables a stock investment policy to be implemented at each stockroom
- Usage profiles provide a range of calculations for forecast usage
- EOQ calculation is performed which could be used as the basis of reorder quantity and time between orders
- Lead time can be factored in the reorder calculation to modify order review periods

Calculation Of Reorder Point, Maximum Stock And EOQs

4/STINT

This procedure calculates the reorder point, maximum stock, and EOQ for all stockroom balance records where it has been requested (See Stockroom Details maintenance).

For each product/stockroom the procedure first of all determines the value/usage category by using the product cost, average usage and the stock weeks matrix. Using the minimum and maximum stockholdings specified for the category, the procedure then calculates the quantities as follows:

Re Order Point

(Minimum Stock Weeks + Lead Time in Weeks)=Average Usage

Maximum Stock

(Maximum Stock Weeks + Lead Time in Weeks)=Average Usage

Economic Order Quantity

$$\sqrt{\frac{\text{2 x Order Placement Cost x Annual Cost}}{\text{Inventory Holding Cost per year}}}$$

This procedure may be requested as an adhoc update process, but would generally be run at period or week ends. There are no selection parameters required - simply selecting the option will submit a job to perform the updates.

The Selection Of The Reorder Policy

Policy 1 (Up to Max)

Maximum stock - Expected + (average usage x lead time)

Policy 2 (ROQ)

Average Usage x lead time x FACTOR percentage

Policy 3(Up to Min)

Calculate EOQ etc. must be set to 0

A ROP may be manually entered
ROP - Expected

The Reorder/Overstock report is then run and reports the recommended order quantity or if there is a stock excess.

For all items, the stock position is reviewed as follows:

Available stock is calculated:

Physical - Frozen - Picked - Allocated=Available

Expected stock is calculated:

Available + Orders + In Transit - Back Order=Expected

- If expected stock is greater than maximum then the product is reported as in EXCESS.
- If expected is less than reorder point, then the product is reported for ORDER action.
- If Order action is required, the reorder quantity is determined. If a reorder quantity has been specified manually, this is recommended, whatever the reorder policy. Otherwise the reorder quantity is determined as follows for each policy:
- If the policy is 'ROP/MAX', the application will recommend reorder up to the maximum stock holding.
- If the policy is 'ROP/ROQ', the application will calculate a reorder quantity based on the Reorder Quantity Factor, the average usage and the lead time.
- If the policy is 'Up to ROP', the application will recommend that you order sufficiently to bring the expected back to ROP.

Examples

Physical/Available = 0

Expected = -90

Max weeks=8 Average Usage = 10

Min weeks=6 Lead Time = 4 weeks

Reorder Factor = 200%

ROP = (Min. weeks) 6 + (Lt) 4 x (av. usage) 10 = 100

Maximum stock = (Max. weeks) 8 x (av.usage) 10 = 80

Maximum stock is below ROP because the lead time (4) is greater than the difference between Maximum (8) and Minimum (6) weeks i.e. 2

Policy 1 (Up to Max)

80 - -90 + 40 = 210

Maximum stock - Expected + (av. usage x lt)

Policy 2 (ROQ)

10 x 4 x 200% = 80

Av. usage x (lt) x Factor

Policy 3 (Reorder on Demand)

0 - -90=90

Reorder Point - Expected

Reorder/Overstock

8/STINR

This procedure produces a report which can be used to identify those products/styles where some action may be necessary due to stock shortage or overstocking, according to the stock management information.

```
IN034              ST - JBA System Test Company     USER GBJBARW2    9/11/95
                                                                     9:50:37
 Request Reorder/Overstock Report

 Sequence      2         (1=Style/Stockroom, 2=Stockroom/Style)
 All Styles?   2         (1=All Styles, 2=Just those Styles that have moved
                          and not yet been reported)













F3=Exit
      05-16     SF        MW        KS        IM        II   KENB20A  PET178S2
```

Fields:

Sequence

Enter the sequence for the report.

1 - Style/Stockroom.

2 - Stockroom /Style.

All Styles ?

1 - All Styles.

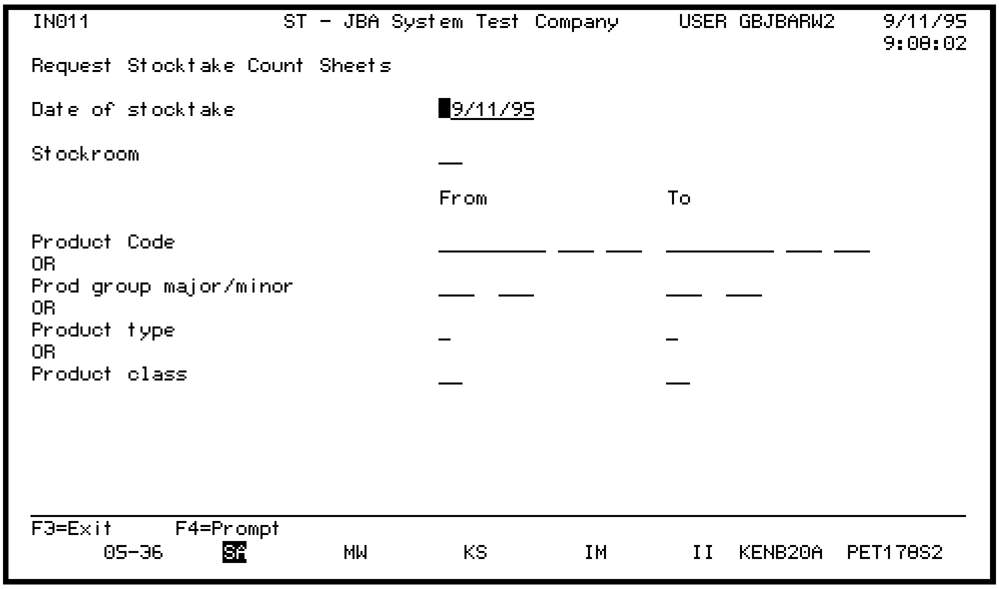
2 - Just the Styles that have moved and not yet been reported.

Enter the requirements for the report and press **Enter** to submit the report.

Request Stocktake Count Sheets Screen

5/STINT

This procedure enables the request of stock count sheets for a specified range of styles or styles, colours sizes and fittings. The computer stock figure for the selected products is recorded at the time of the request.



```
IN011              ST - JBA System Test Company      USER GBJBARW2     9/11/95
                                                                       9:08:02
Request Stocktake Count Sheets

Date of stocktake                  9/11/95

Stockroom                          __

                                   From                To

Product Code                       ________ __ __      ________ __ __
OR
Prod group major/minor             __ __               __ __
OR
Product type                       _                   _
OR
Product class                      __                  __



F3=Exit     F4=Prompt
      05-36     SF         MW        KS        IM        II   KENB20A  PET178S2
```

Fields:

Date Of Stocktake

A valid date to specify the date on which the stocktake is to take place. The date will default to the current date, but it may be changed if required.

Stockroom

This is the stockroom code for the stockroom in which this stocktake is to take place. This field should be left blank if sheets are required for all stockrooms.

Stock Count Sheets can be requested for products in a given range of product number, product group major, product group minor, product type or product class.

If sheets are required for a range then From and To values must be specified. If sheets are only required for a specific product group, for example, then only the product group From need be entered. If all products are required then leave all From and To values blank.

Product Code From/To

This is the range of products to be counted. This can be at colour/size/fitting level if appropriate.

Product Group Major/Minor From/To

Enter the range of product groups required. Either major groups, minor groups or both may be specified.

Product Type From/To

Enter the range of product types required.

Product Class From/To

Enter the range of product classes required.

Once the selections have been made, press **Enter**, this validates the entries made. If you press **Enter** a second time it confirms the selection and submits a job to print the count sheets.

Enter Stock Counts

6/STINT

This procedure enables the entry of details of the stock counts.

This facility allows you to enter stock counts through the following procedural screens:

- Enter Stock Counts - Select.
- Enter Stock Counts - Entry by matrix.
- Enter Stock Counts - Entry by full product.

Enter Stockcounts Screen 1

This screen is displayed when you select the **Enter** Stockcounts option.

It enables the selection of a particular count, by entering the date on which count sheets were requested.

```
IN013                 ST - JBA System Test Company      USER GBJBARW2     9/11/95
                                                                           9:08:50
Enter Stockcounts

Stockroom.......?. _
Stocktake date....  0/00/00











F3=Exit F4=Prompt F13=Full Product

      05-21     SA         MW        KS        IM        II   KENB20A  PET178S2
```

Fields:

Stockroom

The stockroom for which the count is to be entered. A prompt is available.

Stocktake Date

The stocktake date as specified on the count entry sheets

Press **Enter** to confirm the selection. The screen for entry of the stock counts will then be displayed.

Functions:

F13

Full product stock count entry

Enter Stockcounts Item Code/Quantity Selection For Stocktake Window

This screen is displayed when you press **Enter** on the initial screen.

It enables the data entry of physical counts by style, colour, size and fitting following a stock count. The selected stockroom is identified and the nominated stocktake date is displayed.

```
IN013              ST - JBA System Test Company     USER GBJBARW2    9/11/95
                                                                      9:11:09
Enter Stockcounts

Stockroom......... RM  Raw Material Stores
Stocktake date....  8/11/95
Style code......?. PIECE

Item Code/Quantity Selection For Stocktake
  PIECE          piece style        Sighs....................    More:     >
  Color........  B     F     C      1**** D     2**** 3**** A     4**** 8+
  Reddy          _____ _____ _____ _____ _____ _____ _____ _____ _____ _____
  Bluey          _____ _____ _____ _____ _____ _____ _____ _____ _____ _____
  White          _____ _____ _____ _____ _____ _____ _____ _____ _____ _____

F10=Change matrix F12=Previous F20=Window right

      12-19     SA        MW        KS        IM        II   KENB20A   PET178S2
```

Enter the following details for each style counted:

Fields:

Style

The style matrix is displayed showing all applicable colours, sizes and fittings.

Quantity

Enter the total physical stock counted, for each applicable colour, size and fitting combination

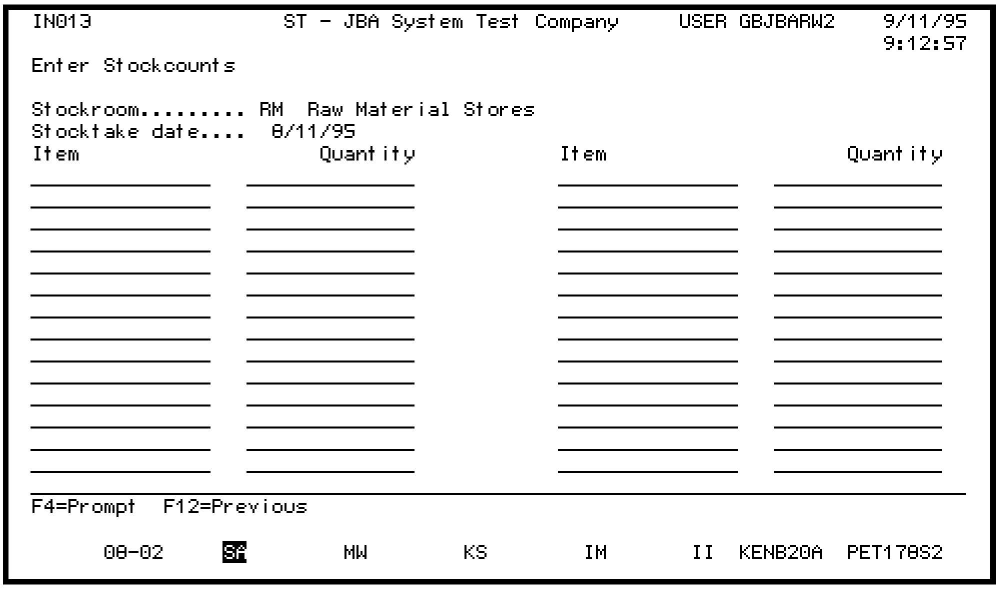
As each screen display is completed the data may be used to update the stockroom balance record by pressing **Enter**.

If an entry has been made incorrectly, or subsequent stock is found which increases the stock on hand, then the complete record should be re-entered, overwriting the previous details.

Enter Stockcounts Screen 2

This screen is displayed when you press **Enter** on the first Enter Stockcounts screen.

It enables the entry of stock counts by full 15 character product code.



```
IN013                ST - JBA System Test Company     USER GBJBARW2    9/11/95
                                                                       9:12:57
Enter Stockcounts

Stockroom......... RM  Raw Material Stores
Stocktake date....  8/11/95
Item                    Quantity             Item                    Quantity

F4=Prompt   F12=Previous

      08-02     SF         MW         KS         IM        II   KENB20A   PET178S2
```

Enter the following details:

Fields:

Item

Enter the item code from the stock count sheet.

Quantity

Enter the total physical stock quantity counted as recorded on the count sheet.

Request Stock Count Reconciliation

7/STINT

There are two objectives for this procedure:

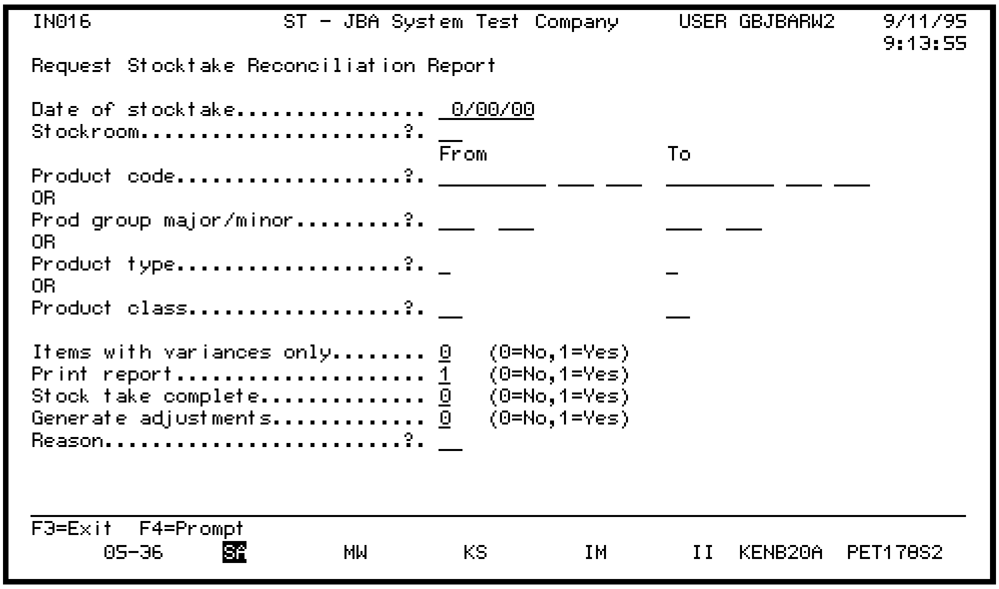
- To provide a comparison of the book stock figure with the actual quantity counted, throughout the stocktake. The report values the physical and counted stock, highlights any discrepancies and values them as appropriate. It should be noted that items subject to FIFO costing are cost based on latest cost for the purposes of the reconciliation report.
- To record the completion of a stocktake and generate, if required, any adjustments to correct the Stockroom Balance Records. Inventory adjustments are costed according to the product's costing method.

The reconciliation is based on the comparison of quantity counted and the recorded computer stock quantity when the stocktake was requested. If automatic adjustment is requested, the current Stockroom Balance Record 'Computer Stock' is adjusted by the difference above, regardless of any stock movements that may have taken place in the interim.

Request Stocktake Reconciliation Report Screen

This screen is displayed when you select the request Stocktake Reconciliation Report option.

It allows you to enter the criteria for the report and submit the report for batch processing.



```
IN016                  ST - JBA System Test Company      USER GBJBARW2     9/11/95
                                                                           9:13:55
Request Stocktake Reconciliation Report

Date of stocktake................  0/00/00
Stockroom......................?.  __
                                  From                 To
Product code...................?.  _________ ___ ___   _________ ___ ___
OR
Prod group major/minor.........?.  ___  ___           ___  ___
OR
Product type...................?.  _                    _
OR
Product class..................?.  __                   __

Items with variances only........  0    (0=No,1=Yes)
Print report.....................  1    (0=No,1=Yes)
Stock take complete..............  0    (0=No,1=Yes)
Generate adjustments.............  0    (0=No,1=Yes)
Reason.........................?.  __


F3=Exit   F4=Prompt
      05-36      SA         MW        KS        IM        II   KENB20A   PET178S2
```

Fields:

Date Of Stocktake

The date on which the stocktake took place.

Stockroom

The stockroom for which this stocktake took place.

The report can be requested for products in a given range of; product number, product group, product type or product class.

Products From/To

The range of products for which a report is required.

Product Group Major/Minor From/To

The range of product groups for which a report is required. You may specify major group, minor group, or both.

Product Type From/To

The range of style types for which a report is required.

Product Class From/To

The range of product classes for which a report is required.

Items With Variance Only

Enter 1 if the report should only show products where there is a variance between the computer stock and the quantity counted, otherwise enter 0 to print all products.

Print Report

Enter 1 if the report is to be printed, otherwise 0. (You may wish to run this option merely to complete the stocktake).

Stocktake Complete

Enter 1 if the stocktake is complete, otherwise 0. The stocktake must be completed if adjustments are to be generated.

Generate Adjustments

Enter 1 if the adjustments are to be automatically generated, otherwise 0. If adjustments are to be generated, a reason code must be entered. This reason code will be associated with the generated adjustment movements.

Reason

Enter a reason code chosen from those defined on the Descriptions File against Major Type MOVR.

Press **Enter** to confirm the selections and submit a batch job to produce the report.

Stocktake Report Selection

8/STINT

This report assists in the reconciliation of physical against counted stock. It can be requested for any product within a stockroom in either a detailed, summary, or transaction type format, within a range of specified dates. It produces a list of appropriate stock movements, providing an audit trail and enabling reconciliation to take place. Reference should be made to the section on 'Reports' .

Stocktake Report Selection Screen

This screen is displayed when you select the Stocktake Report Selection option.

It allows you to select the report criteria and submit the report for processing.

```
IN065                 ST - JBA System Test Company      User GBJBARW2       9/11/95
                                                                            9:14:44
 Stocktaking Report

 Start date .........  0/00/00
 End date ...........  0/00/00
 Product code.....?.. _________ ___ ___
 Stockroom........?.. __
 Detail ............. 0    (0=No,1=Yes)
 Summary ............ 0    (0=No,1=Yes)
 By transaction type. 0    (0=No,1=Yes)

F3=Exit    F4=Prompt
      05-24     SF          MW          KS          IM         II   KENB20A   PET178S2
```

Fields:

Start Date

The date from which the first stock movement record is to be selected.

End Date

The last date for stock movement records to be selected.

Product Code

The style (or colour, size and fitting) for which the report is required. Leave this field blank for all styles.

Stockroom

The stockroom for which the report is required. Leave this blank for all stockrooms.

Detail

Enter 1 to produce the report with full stock movement detail. Enter 0 to produce the report in summary form.

Summary

Enter 1 to produce the report in summary form, otherwise enter 0.

By Transaction Type

Enter 1 to produce the report in summary form sorted by transaction type i.e. customer order issues. Otherwise enter 0.

Press **Enter** to confirm the selections and to submit a job to print the report.